# Uzmar de Jesus Gomez Yanez

## Curriculum Vitae

MSc Scientific Computing and Data Analysis

## Personal Statement

As an ML Programmer at Rockstar Games, I deploy and administer the Kubernetes infrastructure for the ML Animation Team, using tools such as Kubectl, Helm, Ansible, ArgoCD, Grafana, and Hashicorp Vault. I have hands-on experience applying Machine Learning algorithms to complex and realistic problems, using Python and C++ as my main programming languages. I have end-to-end experience with all stages of the Machine Learning model lifecycle, including design, creation, training, deployment, monitoring, and productionalization. I have experience deploying and optimizing Large Language Models (LLMs), including fine-tune for specific use cases and implementing Retrieval-Augmented Generation (RAG) for context-aware responses, creating MCP servers and Langgraph agents, and others.

I hold a Master's degree in Scientific Computing and Data Analysis from Durham University, where I also served as a Student Representative for the MiSCaDA program. I have a strong background in mathematics, physics, and computer science. I am passionate about learning new technologies and techniques, and I enjoy collaborating with other talented and creative professionals.

## Personal Data


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| Github: | https://github.com/uzmargomez |
| ResearchGate: | https://www.researchgate.net/profile/Uzmar-Gomez |


## Education

2022 **MSc in Scientific Computing and Data Analysis**, Department of Computer Science, Durham University, Durham, United Kingdom.

2021 Specialization: Earth and Environmental Sciences
Dissertation: *"GPU Programming with Standard C++"*
Description: Used NVIDIA HPC SDK compiler suite to accelerate ExaHyPE's Finite Volume scheme, a generic code base to simulate wave equations. The resulting code was benchmarked against an OpenMP GPU offloading code port of exactly the same code structure, and we report on some problems and benefits of the new C++ software stack.
Advisor: Dr Tobias Weinzierl

Grade: Distinction | Detailed List of Grades

2018 **BSc in Physics**, Faculty of Science, National Autonomous University of Mexico (UNAM), Mexico City, Mexico.

2011 Dissertation: *"Numerical Study of Vlasov Equation in the Schwarzschild Metric"*
Description: Development of a finite difference scheme for evolving the relativistic Vlasov equation in the context of a black hole metric. This approach assumes the equation is advective, with velocities that vary with both time and position.
Advisor: Dr Miguel Alcubierre

Overall Score: 9.37/10 | Detailed List of Grades

## Computer Skills

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| Programming Languages | Python, C++, Fortran, Julia |
| Machine/Deep Learning | TensorFlow, Keras, PyTorch, ARIMA, SARIMA, LDA, PCA, Kmeans, KNN, Neural Networks, Recommendation systems, Classification and Regression problems |
| Serving | Triton, Seldon Core, TFServing |
| VectorDB | Milvus |
| GenAI | Langgraph, Langfuse, vLLM, LiteLLM, FastMCP |
| Databases | PostgreSQL, Clickhouse, MongoDB, Elasticsearch, Redis |
| Containers | Docker, Kubernetes |
| Operating Systems | GNU/Linux, Windows |
| Web Backend | FastAPI, Flask |
| Web Frontend | Streamlit, HTML (Plotly Dash), Bootstrap |
| Version Control | Git, Perforce |
| Parallel Computing | C++ Standard Algorithms, CUDA Python |
| Data Visualization | Plotly Dash, Qlik, Tableau |
| Automation | Airflow |
| Machine Learning | ClearML, Kubeflow |
| GCP Tools | BigQuery, BQ ML, Compute Engine, Composer (Airflow), Artifact Registry, Kubernetes Engine, Storage, Data Studio, Kubeflow, Data Fusion, Cloud Build, VertexAI |
| Kubernetes Infrastructure | Kubectl, Helm, Kustomize, Ansible, ArgoCD, Grafana, Prometheus, HashiCorp Vault, MetalLB, NGINX Ingress, Istio Gateway, Cilium |

## Experience

*Short Description*

**Mar 2023 - Present**. ML R&D Programmer at Rockstar Games, specializing in the deployment and management of on-premises Kubernetes infrastructure for the ML Animation Team. Proficient in using tools such as Kubectl, Helm, Ansible, ArgoCD, Grafana, and HashiCorp Vault, as well as managing diverse databases including MongoDB, Redis, PostgreSQL and Elasticsearch. Actively involved in all stages of machine learning model development, from data gathering and preprocessing to model training, evaluation, and production deployment. Implemented Large Language Models (LLMs) systems with Langgraph, deployed LLMs with vLLM, integrated Retrieval-Augmented Generation (RAG) systems for context-aware responses, used FastMCP to connect LLMs to tools and data, amongst other tasks.

**Sep 2022 - Mar 2023**. Associate ML R&D Programmer at Rockstar Games.

**Jul 2020 - Jan 2022**. Data Scientist at Rackspace Technology. I helped migrate data from on-premises servers to the GCP cloud. I worked on churn prediction models that used features related to the COVID19 spreading aside from other business-related features to predict customer attrition, this work includes the development of the model using XGBoost, as well as the deployment of it using Kubeflow and the serving via Seldon Core. I was also involved in constructing an SSAS OLAP Cube to be used by the company for inventory-related queries. I worked on NLP tasks, such as extracting themes out of tickets using semantic similarity, by getting embeddings vectors out of sentences with a transformer architecture named Universal Sentence Encoder. Worked on how to translate voice to text. I built two production-level applications using Plotly Dash.

**Dec 2019 - Jun 2020**. Data Scientist at Softtek. I worked on a face recognition system using a method called Sparse Representation and another one using Neural Networks. I acquired a deep understanding of Neural Networks for Face Detection and Recognition, Image Classification, and Language Processing, among others, as well as some experience in Data Visualization tools such as Tableau.

**Sep 2019 - Dec 2019**. Data Scientist Trainee at Softtek. I learned about different statistical and machine learning techniques and algorithms to study a wide variety of problems.

**Sep 2018 - Dec 2019**. I have research experience using the Einstein Toolkit, a software platform created for supporting research in relativistic astrophysics and gravitational physics.

**2016 - 2019**. Worked as a Teacher's Assistant. I've given lectures aimed at Computer Science and Physics Students on the following topics:

- Computer Science.
  https://web.fciencias.unam.mx/asignaturas/102.pdf

- Selected Topics in Relativity, Cosmology and Gravitation 1.
  https://web.fciencias.unam.mx/docencia/horarios/presentacion/295997
- Relativity
  https://web.fciencias.unam.mx/asignaturas/718.pdf
- Mathematics I for Applied Sciences.
  http://www.fciencias.unam.mx/asignaturas/1118.pdf
- Mathematics II for Applied Sciences.
  http://www.fciencias.unam.mx/asignaturas/1216.pdf

## Languages

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| English: | C1 Level - IELTS (2021) |
| Spanish: | Mothertongue |
| French: | Basic |

## References

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|---|---|
| Name: | Dr. Fernando Herrera |
| Occupation: | Senior Data Engineer at Revolut |
| LinkedIn: | https://www.linkedin.com/in/fernando-jose-herrera-elizalde-76a32790/ |

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| Name: | Dr. Charles Mueller |
| Occupation: | Senior Engineer at Amazon |
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| Name: | Dr. Tobias Weinzierl |
| Occupation: | Professor in the Department of Computer Science at Durham University |
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| Name: | Dr. Miguel Alcubierre |
| Institution Name: | Institute of Nuclear Sciences, UNAM. |
| Occupation: | Director, Researcher, Teacher |
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